Factory Operations & Quality Control System: Detailed Standard Operating Requirement Document

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Target System: Complete Shopfloor Quality Assurance, In-Process Tracking & Final Dispatch Clearance System
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1. Executive Summary & Comprehensive Shopfloor Vision

1.1 Operational Goal & Strategic Purpose

This document defines the standard working procedures for the digital quality control, production tracking, and dispatch clearance system built by Zapuza Technologies LLP. The core mission of this system is to eliminate all paper clipboards, loose inspection sheets, hand-written traveler cards, and verbal handoffs across the entire manufacturing floor. By replacing slow and error-prone paper records with rugged, easy-to-use digital tablets, the factory gains total control over product quality and manufacturing speed.

Paper-based systems suffer from illegible handwriting, lost job traveler sheets, delayed supervisor reviews, and unnoticed measurement errors. Bad parts are often machined further or shipped to customers simply because quality records were checked hours after the work was done. The Zapuza system creates an immediate, locked digital boundary at every workstation. It ensures no job lot moves to the next machine, enters heat treatment, or leaves the factory gate unless every dimension, material certificate, process hold time, and supervisor sign-off is 100% complete and verified.

1.2 Core Operational Benefits

- **Zero Defect Escape:** Automated locks prevent out-of-spec components from moving forward in the production line or being packed for shipping.
- **100% Real-Time Shopfloor Visibility:** Supervisors and factory management can view active jobs, floor queues, active machine cycle times, and inspection status live from any web browser.
- **Flawless Part Lineage & Traceability:** Every finished item is permanently linked to its raw material heat lot, mill certificate, CNC program version, machine operator ID, measurement tool calibration number, and supervisor sign-off timestamp.
- **Elimination of Paperwork Bottlenecks:** Quality inspectors record measurements directly into tablet forms with instant pass/fail feedback, removing hours of manual data entry at shift ends.

1.2 Scope of Shopfloor Coverage

The digital system monitors and controls every job lot through fifteen exhaustive shopfloor workflow stages, spanning raw material receiving through final customer delivery:

- **Raw Material Dock Intake:** Logging raw material heat numbers, supplier mill test certificates, lot sizes, and receiving inspection checks before releasing raw stock to the shopfloor.
- **Digital PDI Checklist Setup:** Easy creation of inspection forms containing technical blueprint drawings, visual callouts, measurement tolerances, and required gauge types.
- **Floor Inspector Workflow:** Guided measurement entry on tablets using mandatory station QR code scanning, linear step-by-step locks, and automatic out-of-tolerance warning alerts.
- **Supervisor Review & Rejection Holding Area:** Clear dashboard for shift supervisors to approve finished checklists, issue conditional concessions, or send failing parts to a digital Rejection Holding Area for rework or scrap disposition.
- **Process Aging & Production Time Tracking:** Automatic time tracking for operation start times, active machining times, cooling or chemical curing hold times, and floor queue delays to remove bottlenecks.

- **Final Shipping Gate Clearance:** Automatic gatekeeping that prevents generating shipping gate passes, packing slips, or QR labels unless all production operations and PDI checklists are 100% completed and signed off.
- **Shift Handover & Machine Logs:** Formal digital handovers between outgoing and incoming operators capturing open job status, machine health, tooling wear, and pending quality checks.
- **Implementation & Pilot Run Execution:** Step-by-step rollout plan including baseline audits, champion training, side-by-side paper running, and production go-live sign-offs.

2. Plant Infrastructure Setup (Factories, Workstations, Tooling, Materials)

2.1 Master Digital Factory Structure

To ensure the digital application mirrors the exact physical reality of the factory floor, administrators define a strict master hierarchy within the central system portal before starting production tracking:

1. **Factory Site / Department:** The physical facility or plant floor section (e.g., Main Machining Shop, Heat Treatment Bay, Final Assembly Line).
2. **Machine Tools & Workstations:** Individual production machinery (e.g., CNC Lathe #04, VMC Station #02, Heat Treat Oven #01).
3. **Part Numbers & Manufactured Components:** The physical items being produced on the floor:
    - **Physical Hardware Parts:** Castings, machined shafts, gears, and sub-assemblies requiring dimensional checks.
    - **Machine Tooling & Programs:** Approved CNC programs, fixture setups, and tooling offsets assigned to the part.
    - **Station Maintenance Checks:** Daily machine setup calibration and preventive maintenance sign-offs before running production.
4. **Measuring Gauges & Equipment Register:** Unique digital IDs for every caliper, micrometer, height gauge, plug gauge, and CMM machine, along with current calibration expiry dates.

2.2 Machine & Workstation Calibration Locks

Every production workstation and CNC machine in the system register requires mandatory daily and shift-based setup verification. Before an operator can select a job on their shopfloor tablet, the system checks whether the machine’s daily preventive maintenance (PM) checklist and safety zero-point checks have been completed. If PM checks are overdue, the tablet interface disables job selection for that machine and sends an alert to the maintenance supervisor.

2.3 Inspection Tool Calibration Enforcement

When an inspector performs a dimensional check, the tablet forces them to select or scan the exact gauge serial number being used (e.g., Vernier Caliper #VC-104). The system automatically cross-references the gauge ID against the master calibration register. If the gauge's calibration interval has expired, the tablet blocks the measurement entry, displays an "Expired Tool Calibration" error, and instructs the worker to obtain a calibrated tool from the tool crib.

3. Raw Material Intake & Mill Test Certificate (MTC) Verification

3.1 Dock Receiving & MTC Logging Workflow

The quality trail begins the moment raw materials (bar stock, billets, forgings, or castings) arrive at the plant receiving dock. The dock clerk logs into the system tablet, creates a new Raw Material Intake record, and records the supplier name, purchase order number, material grade, heat number, and physical quantity received.

3.2 Digital MTC Cross-Checking & Lab Sampling

The supplier’s Mill Test Certificate (MTC) must be uploaded to the intake record via tablet camera scan or document upload. Quality assurance personnel review

the certified chemical composition (e.g., Carbon, Manganese, Silicon levels) and mechanical properties (Yield Strength, Tensile Strength, Hardness) against internal engineering specification sheets:

- **Mandatory MTC Verification:** The intake cannot be saved without an attached readable MTC file and verified heat number.
- **Spectro Analysis & Physical Sampling:** If the material grade requires internal lab testing, the system marks the raw lot as "In Lab Testing" and generates sample tagging labels.
- **Physical Barcode Tagging:** Once cleared by the lab, the system prints weather-resistant barcode labels that are attached to each bundle, billet, or plate stack before stock is moved to the raw store room.

3.3 Material Rejection & Supplier Non-Conformance

5. If incoming raw material fails physical dimension checks, spectro chemical tests, or lacks a valid MTC, the dock worker taps "Reject Intake" on the tablet. The system immediately tags the stock as "Supplier Quarantine," notifies Purchasing and Quality Management, and logs a Supplier Non-Conformance Report (SNCR). Quarantine stock cannot be issued to production work orders under any circumstances.

3. Pre-Dispatch Inspection (PDI) Checklist Module

4. PDI Checklist Template Creation & Blueprint Pinning

4.1 Central Template Design Studio

Quality Engineers create Pre-Dispatch Inspection (PDI) and In-Process checklists within the central web portal. Each template is mapped to specific part numbers, operation steps, and customer quality agreements. Templates follow strict version control (e.g., Rev A, Rev B); whenever an engineering drawing changes, a new revision is created, requiring Quality Manager re-approval before shopfloor deployment.

- **Interactive Blueprint Pinning:** Engineers upload standard PDF engineering drawings into the template portal. They place visual, color-coded pins directly on key inspection points (e.g., Pin 1 on Shaft Diameter, Pin 2 on Thread Pitch). On the tablet, inspectors tap a numbered blueprint pin to instantly highlight the matching data field, eliminating confusion about which dimension is being measured.
- **Configuring Inspection Fields:** Templates support all common shopfloor inspection types:
    - *Dimensions with Plus/Minus Tolerances:* Nominal size, upper allowed limit, lower allowed limit, and measurement unit (e.g., Target: 25.00 mm, Upper: +0.05 mm, Lower: -0.02 mm).
    - *Maximum or Minimum Thresholds:* Maximum allowable limits (e.g., Shaft Surface Roughness Ra 0.8 MAX, or Runout 0.02 MAX).
    - *Visual Pass/Fail Checks:* Simple Yes/No toggles for visual inspection (e.g., Free of burrs, deburred edges, rust preventative coating applied).
    - *Inspection Tool Logging:* Mandatory entry or scanning of the exact measuring tool used (e.g., Vernier Caliper #VC-12, Thread Plug Gauge #PG-04) to ensure tool calibration tracking.

4.3 Critical-to-Quality (CTQ) Priority Tagging

Quality Engineers flag vital dimensions as "CTQ" (Critical-to-Quality). CTQ parameters are visually highlighted in orange on floor tablets and carry stricter tolerance rules. Out-of-spec entries on CTQ items instantly freeze the checklist, prevent conditional supervisor concessions, and send immediate email/SMS alerts to the Quality Head.

5. Tablet-Based Shopfloor Inspector Workflow

5.1 Station QR Code Scanning & Physical Presence Rule

1. To prevent quality inspectors from filling out inspection forms from a desk or breakroom, manual keyboard searching for stations or jobs is strictly disabled on inspector tablets. To open an inspection checklist, the worker

MUST physically scan two distinct QR codes using the tablet camera: first, the physical QR code mounted on the machine or workstation frame, and second, the barcode printed on the active Job Traveler Sheet.

2. **Locked Step-by-Step Inspection:** Once an inspection begins, the tablet locks on that job lot. The inspector cannot skip required dimensions, leave fields empty, or switch to another job until all parameters on the checklist are measured and entered.
3. **Instant Tolerance Warnings:** As soon as an inspector types a value, the screen immediately turns GREEN if it is within drawing limits, or RED if it is out of tolerance. Out-of-tolerance entries prompt the tablet to force a defect photo upload using the camera.

5.4 Real-Time Visual Feedback & Mandatory Photo Capture

As values are entered, fields instantly illuminate in bright GREEN for in-tolerance numbers or vivid RED for out-of-tolerance entries. If an out-of-tolerance value is entered, the tablet triggers a mandatory photo capture workflow. The inspector cannot proceed or complete the inspection without taking at least two clear photos of the defect or gauge reading using the tablet camera, providing visual proof of non-conformance.

6. Supervisor Approvals, Concessions, and Quarantine Holding Area

6.1 Shift Supervisor Approval Dashboard

- **Completed inspection forms immediately populate the Shift Supervisor's live approval queue marked as "Pending Review." Supervisors can sort pending reviews by shift, line, machine, or urgency. They inspect submitted dimensions, tool logs, defect photos, and inspector remarks before issuing a final decision.**

- **Automatic Supervisor Notification:** When an inspector finishes a checklist, it moves directly to the Shift Supervisor’s review screen marked as "Awaiting Approval."
- **Supervisor Action Options:**
    - **Approved:** All measurements meet engineering standards. The job lot is cleared for the next process or packing station.
    - **Conditional Concession Release:** If a non-CTQ dimension is slightly out of drawing tolerance but visually or functionally acceptable for customer use, the Quality Manager (or authorized Supervisor) can issue a Concession. The system forces the user to log a detailed engineering justification and select a concession reason code. Concession lots are permanently flagged in quality history.
    - **Quarantine / Rejection Holding Area:** If a job lot is rejected, the system automatically tags the digital traveler as "Quarantined" and updates the physical floor location status to the "Digital Rejection Holding Area." The system prevents all downstream machines from starting work on this job until Material Review Board (MRB) disposition is completed.

6.3 Material Review Board (MRB) Disposition Options

Quarantined lots are reviewed by the Material Review Board (Plant Manager, Quality Head, Production Lead). The MRB executes one of four digital dispositions within the system portal:

- **Rework Order Generation:** Creates a dedicated rework routing traveler specifying exact corrective machining or polishing operations needed to bring the lot back into specification.
- **Scrap Disposition:** Permanently writes off the material, updates inventory levels, logs scrap cost codes, and requires physical verification at the scrap bin station.
- **Return to Vendor (RTV):** Routes raw material back to the dock for vendor return and issues a debit note request.
- **Customer Special Concession:** Holds the batch until formal written customer approval is uploaded to the system record.

7. Process Routing, Travelers, and Complete Part Lineage

7.1 Digital Production Travelers & Process Sequences

When a production work order is released, the system generates a unique Digital Job Traveler. The traveler contains the master process routing sequence (e.g., Op 10: Raw Material Sawing → Op 20: CNC Rough Turning → Op 30: Heat Treatment → Op 40: Precision Grinding → Op 50: Final PDI & Wash → Op 60: Packing). Every operation must be completed and electronically signed off in precise numerical sequence; operators cannot log time or start work on Op 30 if Op 20 is incomplete.

7.2 Complete Part Lineage & Forward/Backward Traceability

The system links raw material intake heat numbers directly to sub-assemblies, work orders, finished part serial numbers, and customer shipping crates. This creates total forward and backward traceability:

- **Forward Recall Traceability:** If a raw material supplier notifies the plant that heat batch #H-8042 had defective steel chemistry, Quality can query

#H-8042 in the portal. In seconds, the system lists every machined part, work order, and customer crate containing steel from that exact heat lot.
- **Backward Field Traceability:** If a end-customer reports a failed component in the field, entering the component's serial number into the system instantly reveals its full manufacturing birth certificate: original heat cert, CNC machine ID, operator name, shift date, exact PDI measurements, calibration IDs of gauges used, and final gate pass clearance.

8. Process Aging, Cooling/Hold Times, and Bottleneck Tracking

8.1 Mandatory Curing, Cooling, and Chemical Hold Times

Certain manufacturing processes require strict physical aging, thermal cooling, or chemical curing hold periods before parts can undergo further machining or testing (e.g., castings resting for 24 hours after stress relief, or painted parts curing for 4 hours). System routings support mandatory "Process Hold Timers."

8.2 Automated Countdown Timers & Interlocks

When a job lot finishes a hold-designated operation (e.g., Op 30 Heat Treatment Discharge), the system starts an automated countdown timer on the digital traveler. If a machinist attempts to scan the job into Op 40 Precision Grinding before the hold timer reaches zero, the tablet screen flashes RED, displays "Mandatory Cooling Hold Active (1 hr 22 min remaining)," and prevents the machinist from starting the operation.

Step-by-step digital traveler tracking across all manufacturing operations:

- **Standard Factory Operations:** Raw Material Cutting, Forging, Rough Machining, Heat Treatment, Precision Grinding, Surface Coating, and Final Assembly.

- **First-Piece Quality Inspection:** Mandatory First-Piece inspection sign-off before a machine operator can run a full shift production lot, plus scheduled hourly frequency checks (e.g., checking 5 parts every 50 produced).

9. First-Piece and Hourly In-Process Inspections

9.1 First-Piece Sign-Off Protocol

Before starting a full production run on any CNC lathe, mill, or assembly station, machine operators must execute a mandatory "First-Piece Quality Inspection." After setting up tools and running the initial part, the operator and QA inspector measure all key setup dimensions on the tablet interface. The system locks the station in "Setup Mode" until the QA inspector signs off on the first piece. Production counters cannot increment until first-piece clearance is granted.

9.2 Frequency-Based Hourly Inspection Prompts

During active production runs, the system enforces periodic hourly sampling (e.g., check 3 parts every 50 parts produced, or check every 60 minutes). When an inspection interval is reached, the tablet emits an audible alert and displays a visual banner: "Periodic Quality Inspection Due." The operator has a 15-minute grace period to log the required sample measurements before the tablet temporarily pauses job timer tracking.

- **Automated Operation Timestamps:** Records exact clock-in and clock-out timestamps, active machine cycle runtimes, operator ID, and shift number for every operation step.
- **Process Age & Hold-Time Alerts:** Calculates total elapsed processing time and enforces mandatory hold or cooling times (e.g., verifying a batch rested for 4 hours after heat treatment before grinding).

8.3 Live Shopfloor Bottleneck & Queue Monitoring

Because operators scan job travelers in and out at every station, the central portal tracks total elapsed time, active machining time, and idle queue time for every job lot. Supervisors monitor live heatmaps showing floor bottlenecks, work orders stuck in queue between stations for too long, or machines running significantly slower than target cycle times.

10. Pre-Dispatch Clearance Gates and Shipping Interlocks

- ### 5.1 Automatic Pre-Dispatch Quality Check
  - **Automated Shipping Interlock:** Before packing or dispatch clearance, the system automatically checks three mandatory rules:
    a. 100% of all required machining and manufacturing operations are signed off as complete.
    b. All process aging and hold-time requirements have been fulfilled without outstanding quality holds.
    c. Final Pre-Dispatch Inspection (PDI) checklist is signed and Approved by Quality.
  - **Locked Shipping Gate Pass:** Packing slips, dispatch manifests, and gate passes are strictly locked until all three conditions above are verified green.

10.2 Shipping Interlock Logic & Gate Pass Lockouts

If any of the three conditions fail—for instance, if an intermediate deburring operation was skipped, or if the supervisor review is still pending—the central system applies an electronic "Shipping Interlock." Warehouse clerks attempting to print a dispatch gate pass, packing slip, or commercial invoice receive a high-visibility lockout warning detailing the exact unfulfilled requirement.

- ### 11. Dispatch Manifests and Customer QR Labels
  - **Shipping Manifest Logging:** Logs customer Purchase Order (PO) number, batch size, transport vehicle details, and assigned serial numbers.

- **Printable Customer Verification Labels:** Generates shipping QR labels for crate exteriors. Customers can scan the QR label to instantly view the certified digital inspection report for that shipment.

11.3 Digital Inspection Certificates for End-Customers

Scanning the crate QR code opens a secure web document displaying the complete certified quality report for that specific batch: nominal specs, actual measured values, lab heat certs, supplier MTCs, and authorized Quality Lead signatures. This provides customers with complete transparency and eliminates paper Certificate of Analysis (CoA) mailings.

12. Detailed Role-Based Responsibilities & Shift Handover Logs

Plant User Role	Main Responsibilities	System Permissions	Quality Control Safeguards
Plant / General Manager	Overall factory performance oversight, system settings, plant reports	Full view of all plant reports, user setup, high-level overrides	Cannot alter past logged inspection values; all actions tracked
Quality Engineer	Drawing setup, inspection template creation, tolerance definitions	Upload blueprints, pin inspection points, set measurement limits	Cannot alter templates once released to active production
Quality Manager	Review and approve new inspection templates before shopfloor deployment	Approve or reject template setups, issue official concessions	Cannot edit floor measurements; enforces strict quality standards

Plant User Role	Main Responsibilities	System Permissions	Quality Control Safeguards
Shift Supervisor	Shopfloor oversight, review completed checklists, handle rejected lots	Sign off PDI forms, manage Rejection Holding Area, clear lots for dispatch	All lot approvals and rejection moves are permanently logged
QA Floor Inspector	Conduct floor inspections on shopfloor tablets	Scan station QR codes, type dimensional checks, capture defect photos	Locked to station; cannot edit submitted inspection data

12.2 Digital Shift Handover Logbook

At shift changes (e.g., Morning to Evening Shift), outgoing machine operators and shift supervisors complete a mandatory Digital Shift Handover form on their station tablets. The log records: active job status, total parts produced, current scrap count, open quality alerts, pending PDI reviews, tool wear status, and machine health notes. Incoming workers must review and digitally acknowledge the handover log before starting work, ensuring flawless

Plant User Role	Main Responsibilities	System Permissions	Quality Control Safeguards
			communication across shift changes.

13. Factory Hardware, Glove-Friendly UI, Offline Caching, and Security Audit Trails

- ### 7.1 System Security & Audit Trail
  - **Secure Factory Data:** All inspection reports and customer quality records are encrypted and protected from unauthorized external access.
  - **Tamper-Proof QR Labels:** Station and shipping QR codes use secure digital signatures to prevent fake label printing.
  - **Permanent Activity Log:** Every inspector measurement entry, supervisor sign-off, and quality rejection is permanently recorded with user ID, date, and exact time.

- ### 7.2 Shopfloor Ergonomics & Offline Operation
  - **Glove-Friendly Touch Screen UI:** High-contrast screens with large buttons designed specifically for shopfloor industrial tablets used by workers wearing gloves.
  - **Automatic Offline Storage:** If factory Wi-Fi drops, tablets store inspection data locally. Once Wi-Fi reconnects, saved inspection records automatically sync back to the main server without losing work.

13.3 Tamper-Proof Audit Trail & Data Security

Every button tap, measurement entry, supervisor sign-off, concession grant, and blueprint pin update is permanently recorded in an immutable, tamper-proof system audit log. The audit log records: User ID, exact timestamp, IP address, device serial number, previous value, and new value. Past inspection data cannot

be altered or deleted by any user level, ensuring complete compliance during ISO, aerospace, or automotive customer quality audits.

8. Deliverables Included by Zapuza Technologies LLP

14. Implementation Phasing, Training, and Pilot Run Guidelines

To ensure smooth shopfloor adoption without disturbing ongoing production, implementation follows a structured four-phase roadmap over an eight-week timeline:

- **Phase 1: Infrastructure Setup & Master Data Configuration (Weeks 1-2):** Installing industrial Wi-Fi access points, mounting workstation QR plates, setting up central web portal, registering machines, and configuring part templates/blueprints.
- **Phase 2: Champion Training & Dry-Run Testing (Weeks 3-4):** Intensive hands-on training for Quality Engineers, Supervisors, and selected Shopfloor Champions using sample work orders in a controlled lab environment.
- **Phase 3: Pilot Line Parallel Execution (Weeks 5-6):** Deploying system tablets to one designated pilot production line (e.g., CNC Machining Line #1). Running paper travelers and digital tablets side-by-side to validate measurement accuracy and operator speed.
- **Phase 4: Full Factory Go-Live & Paper Decommissioning (Weeks 7-8):** Rolling out tablets across all shopfloor departments, activating shipping clearance gate interlocks, and removing all physical paper inspection clipboards.
- **Central Management Web Portal:** Master system for plant machine registry, drawing template builder, user permissions, supervisor approvals, Rejection Holding Area, and shopfloor reports.
- **Inspector Tablet Application:** Easy-to-use tablet app featuring QR station scanning, locked checklist entry, defect photo capture, and offline saving.
- **Material History & Process Hold Engine:** Raw material heat lot tracing, stage-by-stage operation tracking, cooling/hold-time verification, and bottleneck dashboards.

- **Dispatch Clearance & Shipping Label Module:** Automatic pre-dispatch gatekeeping lock, packing list generation, and customer verification shipping QR label printer support.